Name: Isha Verma Roll No: 21CS101
Mid-Term answer sheet

Q1. Overfitting means the model memorises the training data, including the noise in it, so training accuracy is very high but it does badly on new test data - it does not generalise. Two ways to reduce it:
(i) Dropout - randomly switching off neurons during training so the network cannot rely on specific neurons and learns more robust features.
(ii) Early stopping - we watch the validation loss and stop training when it starts going up, before the model starts memorising.

Q2. In self attention every token embedding is multiplied by three learned matrices Wq, Wk, Wv to get query, key and value vectors. To find how much token i should attend to token j we take the dot product of q_i with k_j. Doing this for all tokens gives the matrix QK^T. Then we divide by sqrt(dk) and apply softmax row wise so each row of weights adds to 1. Finally the output for each token is the weighted sum of the value vectors, softmax(QK^T/sqrt(dk))V. We scale because when dk is large the dot products become large in magnitude, the softmax then becomes very sharp (almost one-hot) and its gradients become very small, which makes training slow and unstable. Dividing by sqrt(dk) keeps the variance of the scores around 1.

Q3. Precision = TP/(TP+FP) = 40/50 = 0.8
Recall = TP/(TP+FN) = 40/60 = 0.667
F1 = 2PR/(P+R) = 2(0.8)(0.667)/(1.467) = 1.067/1.467 = 0.727
Accuracy = (TP+TN)/N = (40+30)/100 = 0.70

(continued)

Q4. (a) GAN: two networks, generator and discriminator, play a minimax game. Discriminator classifies real vs fake, generator tries to fool it. Diffusion: noise is added step by step to the image (forward process) and a neural network (usually a U-Net) is trained to predict the noise so it can reverse the process.
(b) GANs are unstable to train - the generator and discriminator have to stay balanced, and training can oscillate. Diffusion models use a simple MSE loss on the noise so training is stable.
(c) GANs suffer from mode collapse where the generator only produces a few kinds of outputs. Diffusion models give more diverse samples.
(d) GANs are fast since one forward pass produces an image. Diffusion is slow because it needs many denoising steps, although DDIM and distillation reduce the steps.

Q5. RAG pipeline: first the documents are broken into chunks and each chunk is converted into an embedding using an embedding model and stored in a vector database. When a user asks a question, the question is embedded too and we do a similarity search (cosine) to get the top-k chunks. These chunks are added to the prompt with the question and the LLM writes an answer based on them.
Failure 1: the retriever brings back chunks that are not relevant (e.g. the question uses different words than the document). Fix: use hybrid search (BM25 + embeddings) and a re-ranker model on the top results.
Failure 2: the LLM still hallucinates and adds facts that are not in the retrieved text.
Fix: tell the model to answer only from the context and say "I don't know" otherwise, and ask it to cite which chunk each claim comes from.

Name: Dev Patel Roll No: 21CS102
Mid-Term answer sheet

Q1) Overfitting is when the model is too complex and gets high accuracy on training data but low accuracy on testing data.
To reduce it we can use regularization. Another method is to use more data.

Q2) Self attention lets each word look at the other words in the sentence. We create query key and value. The query is multiplied with the keys to get scores and these scores tell how important each word is. Then softmax is applied to make them probabilities. The values are then multiplied by these probabilities and added up to get the output.
We divide by root of dk to make the computation faster.

Q3) Precision = TP / (TP + FN) = 40/60 = 0.67
Recall = TP / (TP + FP) = 40/50 = 0.8
F1 = 2 x 0.67 x 0.8 / (0.67 + 0.8) = 0.729
Accuracy = 40+30 / 100 = 0.7

(continued)

Q4) GANs have a generator and a discriminator which compete with each other. Diffusion models add noise to images and then learn to remove it.
GANs are harder to train than diffusion models. GANs have mode collapse problem.
Diffusion models are better quality but slower.

Q5) In RAG we retrieve relevant documents from a database and give them to the LLM so it can answer questions with up to date information. The documents are stored as embeddings.
One problem is hallucination. We can solve it by using a bigger model.
Another problem is the retrieval can be slow when there are many documents.

Name: Nikhil Rao Roll No: 21CS103
Mid-Term answer sheet

Ans 3.
precision = 40/(40+10) = 0.8
recall = 40/(40+20) = 0.75
F1 = 2*0.8*0.75/(0.8+0.75) = 1.2/1.55 = 0.77

Ans 1.
Overfitting is when the model does not learn enough from the data and the accuracy is low on both training and test data. We can fix it by training for more epochs.

Ans 4.
GAN uses generator and discriminator. Diffusion model uses a transformer to generate images from text prompt like DALL-E. Diffusion is better than GAN.

(continued)

Ans 2.
attention = softmax(QK^T/sqrt(dk))V
Q is query K is key V is value.

(Q5 not attempted - ran out of time)

Name: Tanvi Shah Roll No: 21CS104
Mid-Term answer sheet

Q1. Overfitting happens when a model learns the training examples too well, including random noise, so it shows very low training error but high error on validation/test data.
1) L1/L2 regularization - adds a penalty for large weights so the model becomes simpler.
2) Data augmentation - creating more varied training examples (flips, crops) so the model can't just memorise them.

Q2. Each token gets Q, K and V vectors by multiplying its embedding with weight matrices that are learned in training. Score = Q.K for every pair of tokens, then softmax converts the scores into weights that sum to one, and the output is the weighted sum of the V vectors. We scale by sqrt(dk) to keep the numbers small.

Q3. P = 40/(40+10) = 0.8
R = 40/(40+20) = 0.67
F1 = 2*0.8*0.67/(0.8+0.67) = 0.73
Accuracy = (40+30)/100 = 0.7

(continued)

Q4. GAN: a generator and a discriminator play a minimax game. Diffusion: the model learns to denoise images that were corrupted with Gaussian noise over many timesteps.
GAN training is unstable and can have mode collapse; diffusion training is stable and the samples are more diverse.
GANs are fast (a single forward pass), diffusion is slow (hundreds of denoising steps).

Q5. RAG: chunk the documents, embed each chunk with an embedding model and store the vectors in a vector DB like FAISS. At query time embed the question, retrieve the top-k similar chunks and put them in the prompt for the LLM.
Failure: chunks too large or too small, so the right information is split or lost -> tune the chunk size and use overlap between chunks.
Another issue is cost.

Name: Farhan Ali Roll No: 21CS105
Mid-Term answer sheet

Q1. Overfitting: the model performs well on training data but fails to generalise to unseen data. We can use dropout and cross validation.

Q2. Self-attention computes attention between all words. Formula: Attention(Q,K,V) = softmax(QK^T/sqrt(dk))V where Q, K, V come from the input through linear layers. The softmax gives the weights. Scaling stops the softmax from saturating and having vanishing gradients when dk is large.

Q3. Precision = 40/50 = 0.8, Recall = 40/60 = 0.67, F1 = (0.8+0.67)/2 = 0.735, Accuracy = 70%

(continued)

Q4. GANs: two models trained against each other. Diffusion models: a forward noising process and a reverse denoising process. GANs are faster at generation. Diffusion models are more stable to train and produce higher diversity.

Q5. RAG = retrieval + generation. Steps: user query -> search the knowledge base -> retrieved text + query given to the LLM -> answer.
Failure 1: wrong documents retrieved -> use re-ranking.
Failure 2: outdated knowledge base -> regularly update the index.

Name: Pooja Nair Roll No: 21CS106
Mid-Term answer sheet

Q1. Overfitting - training accuracy high, test accuracy low; the model learnt the noise.
Methods: regularisation (penalise big weights), getting more training data.

Q2. Q = XWq, K = XWk, V = XWv.
Attention = softmax(QK^T / sqrt(dk)) V.
Softmax makes the weights between 0 and 1.

Q3. P = 0.8, R = 0.667, F1 = 2(0.8)(0.667)/(1.467) = 0.727, acc = 0.7

(continued)

Q4. (a) GAN adversarial, diffusion denoising. (b) GANs unstable. (c) GAN mode collapse. (d) GAN faster.

Q5. Index the docs as embeddings; retrieve the similar chunks for a query; the LLM generates the answer from them.
Failure: hallucination - prompt it to only use the context.